High school · counting

Paths, combinations and Pascal

Worked solutions

Exercise 1

Find C(8,2) and explain why it equals C(8,6).

$C(8,2) = 8 \times \frac{7}{2} = 28$. Choosing two excluded objects is equivalent to choosing six included objects.

Exercise 2

Count monotone paths from (0,0) to (4,3).

Choose four rightward steps among seven: $C(7,4) = 35$.

Exercise 3

How many of these paths pass through (2,1)?

$C(3,2) \times C(4,2) = 3 \times 6 = 18$.

Exercise 4

Choose a team of three from eight people. Two specified people cannot be together. How many teams remain possible?

There are $C(8,3) = 56$ teams. Exactly 6 contain both specified people. Therefore 50 teams remain.

Problem: complete reasoning

Count monotone paths from (0,0) to (5,4) that avoid A(2,1) and B(3,3). Count all paths, then those passing through A, through B and through both. Explain why the last category must be added back.

Total: $C(9,5) = 126$. Through A: $C(3,2)C(6,3) = 60$. Through B: $C(6,3)C(3,2) = 60$. Through both: $3 \times C(3,1) \times 3 = 27$. Paths through both were subtracted twice; adding them back once gives $126 - 60 - 60 + 27 = 33$.